

15.10 Solution: This problem can be solved in the same way as Problem 14.22, and the difference is that, there is no fundamental frequency for non-periodic motion.

(a) Consider the Fourier transform of the velocity. Similar to Problem 14.22,

$$\begin{aligned} \mathbf{v}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{i\omega t} \dot{\mathbf{r}}(t) dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp \left\{ \frac{i\omega}{\omega_0} (\xi + \epsilon \sinh \xi) \right\} \begin{pmatrix} a \sinh \xi \\ -b \cosh \xi \end{pmatrix} d\xi. \end{aligned}$$

Make the integration variable change, $\xi \rightarrow \xi + i\pi/2$, and with $\sinh \xi \rightarrow i \cosh \xi$, the x -component can be evaluated as

$$v_x(\omega) = \frac{iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp \left\{ \frac{i\omega}{\omega_0} \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \cosh \xi d\xi.$$

Using the following identity,

$$K_\nu(z) = \int_0^{+\infty} e^{-z \cosh t} \cosh(\nu t) dt,$$

we can express the above integral as

$$\begin{aligned} v_x(\omega) &= \frac{iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \frac{d}{d(\omega\epsilon/\omega_0)} \int_{-\infty}^{+\infty} \exp \left\{ \frac{i\omega}{\omega_0} \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \cosh \xi d\xi \\ &= \frac{iae^{-\omega\pi/2\omega_0}}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left(\exp \left\{ \left(\frac{i\omega}{\omega_0} + 1 \right) \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \right. \\ &\quad \left. + \exp \left\{ \left(\frac{i\omega}{\omega_0} - 1 \right) \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \right) d\xi \\ &= \frac{iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \int_0^{+\infty} \left(\exp \left\{ -\frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \cosh \left(\left(\frac{i\omega}{\omega_0} + 1 \right) \xi \right) \right. \\ &\quad \left. - \exp \left\{ -\frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \cosh \left(\left(\frac{i\omega}{\omega_0} - 1 \right) \xi \right) \right) d\xi \\ &= \frac{iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \left[K_{i\omega/\omega_0+1} \left(\frac{\omega\epsilon}{\omega_0} \right) + K_{i\omega/\omega_0-1} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]. \end{aligned}$$

For modified Bessel function, we have the recurrence relation

$$K_{\nu+1}(z) + K_{\nu-1}(z) = 2K'_\nu(z),$$

and

$$v_x(\omega) = \frac{2iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right).$$

Therefore, the Fourier transform of $x(t)$ is

$$x(\omega) = \frac{i}{\omega} v_x(\omega) = -\frac{2ae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}\omega} K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right).$$

Similarly, for the y -component,

$$v_y(\omega) = -\frac{ibe^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \left[K_{i\omega/\omega_0+1} \left(\frac{\omega\epsilon}{\omega_0} \right) - K_{i\omega/\omega_0-1} \left(\frac{\omega\epsilon}{\omega_0} \right) \right].$$

Using the recurrence relation

$$K_{\nu+1}(z) - K_{\nu-1}(z) = \frac{2\nu}{z} K_{\nu}(z),$$

we have

$$v_y(\omega) = \frac{2be^{-\omega\pi/2\omega_0}}{\sqrt{2\pi\epsilon}} K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right),$$

and

$$y(\omega) = \frac{i}{\omega} v_y(\omega) = \frac{2ib e^{-\omega\pi/2\omega_0}}{\sqrt{2\pi\omega\epsilon}} K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right).$$

Then, from the dipole approximation,

$$\frac{dI}{d\omega} I(\omega, b) = \frac{4}{3c^3} |\ddot{\mathbf{d}}(\omega)|^2 = \frac{4\omega^4}{3c^3} |\mathbf{d}(\omega)|^2,$$

where $\mathbf{d}(\omega) = ze(x(\omega), y(\omega))$, we will have the final result

$$\begin{aligned} \frac{d}{d\omega} I(\omega, b) &= \frac{4\omega^4}{3c^3} \left[\frac{2a^2 z^2 e^2 e^{-\pi\omega/\omega_0}}{\pi\omega^2} \left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{2b^2 z^2 e^2 e^{-\pi\omega/\omega_0}}{\pi\omega^2 \epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] \\ &= \frac{8}{3\pi} \frac{(ze a \omega)^2}{c^3} e^{-\pi\omega/\omega_0} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{b^2/a^2}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] \\ &= \frac{8}{3\pi} \frac{(ze a \omega)^2}{c^3} e^{-\pi\omega/\omega_0} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{\epsilon^2 - 1}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right]. \end{aligned}$$

(b) From part (a), the radiation cross section is

$$\begin{aligned} \frac{d\chi(\omega)}{d\omega} &= 2\pi \int_0^{+\infty} \frac{d}{d\omega} I(\omega, b) b db \\ &= \frac{16}{3} \frac{(ze a \omega)^2}{c^3} e^{-\pi\omega/\omega_0} \int_0^{+\infty} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{\epsilon^2 - 1}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] b db. \end{aligned}$$

Since $\epsilon = \sqrt{1 + (b/a)^2}$, then $b = a\sqrt{\epsilon^2 - 1}$, $db = a\epsilon d\epsilon/\sqrt{\epsilon^2 - 1}$, and $b db = a^2 \epsilon d\epsilon$. The integral becomes

$$a^2 \int_1^{+\infty} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{\epsilon^2 - 1}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] \epsilon d\epsilon,$$

which, upon another variable change, $\epsilon = (\omega_0/\omega)x$, becomes

$$a^2 \left(\frac{\omega_0}{\omega} \right)^2 \int_{\omega/\omega_0}^{+\infty} \left[\left[K'_{i\omega/\omega_0}(x) \right]^2 + \left(1 - \frac{\omega^2/\omega_0^2}{x^2} \right) \left[K_{i\omega/\omega_0}(x) \right]^2 \right] x dx.$$

The first term can be evaluated with integration by parts,

$$\begin{aligned} \int_{\omega/\omega_0}^{+\infty} x \left[K'_{i\omega/\omega_0}(x) \right]^2 dx &= \int_{\omega/\omega_0}^{+\infty} x K'_{i\omega/\omega_0}(x) d(K_{i\omega/\omega_0}(x)) \\ &= x K_{i\omega/\omega_0}(x) K'_{i\omega/\omega_0}(x) \Big|_{\omega/\omega_0}^{+\infty} - \int_{\omega/\omega_0}^{+\infty} K_{i\omega/\omega_0}(x) d(x K'_{i\omega/\omega_0}(x)). \end{aligned}$$

The modified Bessel function of the second kind satisfies the following ODE,

$$\frac{d}{dx} [xK'_\nu(x)] = \left(1 + \frac{\nu^2}{x^2}\right) K_\nu(x),$$

then we are left with the boundary term only,

$$\frac{d\chi(\omega)}{d\omega} = \frac{16}{3} \frac{(zeav)^2}{c^3} e^{-\pi\omega/\omega_0} a^2 \left(\frac{\omega_0}{\omega}\right)^2 \left[xK_{i\omega/\omega_0}(x)K'_{i\omega/\omega_0}(x) \right] \Big|_{\omega/\omega_0}^{+\infty}.$$

Since $K_\nu(x)$ vanishes exponentially at infinity, and notice that $\omega_0 = v/a$, we are finally left with

$$\frac{d\chi(\omega)}{d\omega} = \frac{16}{3} \frac{(zeav)^2}{c^3} e^{-\pi\omega/\omega_0} \frac{\omega}{\omega_0} K_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0}\right) \left[-K'_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0}\right) \right].$$

(c) For $\omega/\omega_0 \ll 1$, using the asymptotic form of the Bessel function,

$$K_\nu(x) \rightarrow -\log\left(\frac{x}{2}\right), \quad K'_\nu(x) \rightarrow -\frac{1}{x},$$

we can write the cross section as

$$\frac{d\chi(\omega)}{d\omega} = \frac{16}{3} \frac{(zeav)^2}{c^3} \frac{\omega}{\omega_0} \left[-\log\left(\frac{\omega}{2\omega_0}\right) \right] \left(\frac{\omega_0}{\omega}\right) = \frac{16}{3} \frac{(zeav)^2}{c^3} \log\left(\frac{2\omega_0}{\omega}\right).$$

Since $a = Ze^2/mv^2$, we then have

$$\begin{aligned} \frac{d\chi(\omega)}{d\omega} &= \frac{16}{3} \frac{(Zz^2e^3/mv)^2}{c^3} \log\left(\frac{2mv^3}{Zze^2\omega}\right) \\ &= \frac{16}{3} \frac{Z^2e^2}{c^3} \left(\frac{z^2e^2}{m}\right)^2 \frac{1}{(\beta c)^2} \log\left(\frac{2mv^3}{Zze^2\omega}\right) \\ &= \frac{16}{3} \frac{Z^2e^2}{c} \left(\frac{z^2e^2}{mc^2}\right)^2 \frac{1}{\beta^2} \log\left(\frac{2mv^3}{Zze^2\omega}\right), \end{aligned}$$

which agrees with Eq. (15.25), up to some constant factor.

For $\omega/\omega_0 \gg 1$, using the asymptotic result $K_\nu(x) \rightarrow e^{-x} \sqrt{\pi/2x}$, we can see that the limiting form is proportional to $e^{-(\pi+2)\omega/\omega_0}$.

(d)